12.14.4 Esters

Methods of preparation

1. Esterification

We have already learnt that treatment of alcohols with carboxylic acids in presence of mineral acid gives esters. The reaction is carried to completion by using an excess of reactant or by removing the water from the reaction mixture.

2. Alcoholysis of Acid chloride or Acid anhydrides

ii) Treatment of acid chloride or acid anhydride with alcohol also gives esters

Physical Properties

Esters are colour less liquids or solids with characteristic fruity smell. Flavours of some of the esters are given below.

S.No	Ester	Flavour
1	Amyl acetate	Banana
2	Ethyl butyrate	Pineapple
3	Octyl acetate	Orange
4	Isobutyl formate	Raspberry
5	Amyl butyrate	Apricot

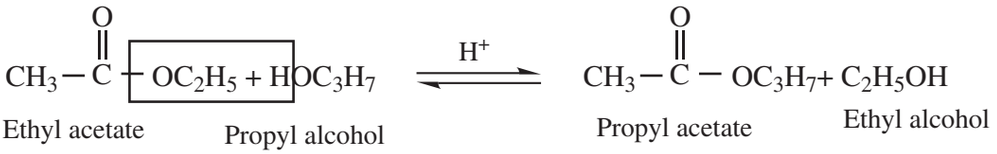
Chemical Properties

1. Hydrolysis

We have already learnt that hydrolysis of esters gives alcohol and carboxylic acid.

2. Reaction with alcohol (Transesterification)

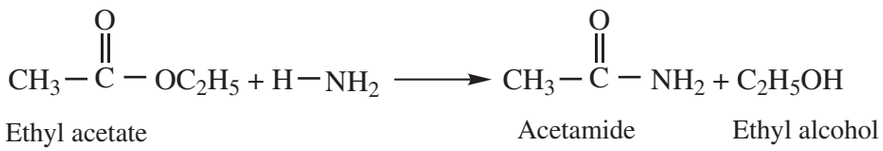
Esters of an alcohol can react with another alcohol in the presence of a mineral acid to give the ester of second alcohol. The interchange of alcohol portions of the esters is termed **transesterification**



The reaction is generally used for the preparation of the esters of a higher alcohol from that of a lower alcohol.

3. Reaction with ammonia (Ammonolysis)

Esters react slowly with ammonia to form amides and alcohol.



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